# Data Analysis Skills – Group Project 2 Datasets Allocations

**Datasets**

There are 25 datasets from which your group has been allocated one to work on this assignment. The dataset number for your group is the same number as your group (e.g. group 10 will use dataset 10). Please ensure you are working with the correct dataset and are answering your assigned question of interest. The datasets are available for you to download from Moodle in "Week 9-11 – Group Project 2"

**Datasets 1- 5**

Datasets 1-5 come from the FIES (Family Income and Expenditure Survey) recorded in the Philippines. The survey, which is undertaken every three years, is aimed at providing data on family income and expenditure. Each of the 5 datasets obtained from this survey are from different regions across the Philippines. You will have access to the following variables, recorded by household (Note: head of the household refers to the person "in charge" of the house):

- `Total.Household.Income` – Annual household income (in Philippine peso)
- `Region` – The region of the Philippines which you have data for
- `Total.Food.Expenditure` – Annual expenditure by the household on food (in Philippine peso)
- `Household.Head.Sex` – Head of the households sex
- `Household.Head.Age` – Head of the households age (in years)
- `Type.of.Household` – Relationship between the group of people living in the house
- `Total.Number.of.Family.members` – Number of people living in the house
- `House.Floor.Area` – Floor area of the house (in $m^2$)
- `House.Age` – Age of the building (in years)
- `Number.of.bedrooms` – Number of bedrooms in the house
- `Electricity` – Does the house have electricity? (1=Yes, 0=No)

**Task**

Imagine you have been asked by government officials to investigate the following question of interest:

- Which household related variables influence the number of people living in a household?

You should conduct an analysis to answer your question using a Generalised Linear Model (GLM). Following your analyses, you should then summarise your results in the form of a presentation.

**Datasets 6 – 10**

Datasets 6-10 come from the IMDB film database. The database contains a variety of information on all films that have been released. You will work with a subset of this database and will have access to the following variables, recorded by film:

- `film.id` – The unique identifier for the film
- `year` – Year of release of the film in cinemas
- `length` – Duration (in minutes)
- `budget` – Budget for the films production (in $1000000s)
- `votes` – Number of positive votes received by viewers
- `genre` – Genre of the film
- `rating` – IMDB rating from 0-10

**Task**

Imagine you have been asked by a film producer to investigate the following question of interest:

- Which properties of films influence whether they are rated by IMDB as greater than 7 or not?

You should conduct an analysis to answer your question using a Generalised Linear Model (GLM). Following your analyses, you should then summarise your results in the form of a presentation.

**Datasets 10 – 15**

Datasets 10-15 come from the Coffee Quality Database (CQD). The database contains information from the Coffee Quality Institute which is a non-profit organisation working internationally to improve the quality of coffee and the lives of the people who produce it. Each of the 5 datasets contain information on features of coffee and its production, including an overall score of quality. You will have access to the following variables, recorded by batch.

- `country_of_origin` – Country where the coffee bean originates from.
- `aroma` – Aroma grade (ranging from 0-10)
- `flavor` – Flavour grade (ranging from 0-10)
- `acidity` – Acidity grade (ranging from 0-10)
- `category_two_defects` – Count of category 2 type defects in the batch of coffee beans tested.
- `altitiude_mean_meters` – Mean altitude of the growers farm (in metres)
- `harvested` – Year the batch was harvested
- `Qualityclass` – Quality score for the batch (Good - $\geq 82.5$, Poor - <82.5). Note: 82.5 was selected as the cut off as this is the median score for all the batches tested.

**Task**

Imagine you have been asked by a local coffee farmer to investigate the following question of interest:

- What influence do different features of coffee have on whether the quality of a batch of coffee is classified as good or poor?

You should conduct an analysis to answer your question using a Generalised Linear Model (GLM). Following your analyses, you should then summarise your results in the form of a presentation.

**Datasets 16-20**

Datasets 16-20 come from the Dallas animal shelter. Each of the 5 datasets contain a variety of information relating to each animal admitted to the shelter. You will have access to the following variables, recorded by animal admission:

- Animal_type – The type of animal admitted to the shelter
- Month – Month the animal was admitted, recorded numerically with January=1
- Year. – Year the animal was admitted to the shelter.
- Intake_type – Reason for the animal being admitted to the shelter
- Outcome_type – Final outcome for the admitted animal
- Chip_Status – Did the animal have a microchip with owner information?
- Time_at_Shelter – Days spent at the shelter between being admitted and the final outcome.

**Task**

Imagine you have been asked by the shelter management to investigate the following question of interest:

- Which factors influence the number of days an animal spends in the shelter before its final outcome is decided?

You should conduct an analysis to answer your question using a Generalised Linear Model (GLM). Following your analyses, you should then summarise your results in the form of a presentation.

**Datasets 21-25**

Datasets 21-25 were collected at Hewlett-Packard Labs and shared with the UCI Machine Learning Repository. Each of these 5 datasets contain data on classifying emails as spam or not spam, with additional variables indicating the frequency of certain words and characters in the email. You will have access to the following variables:

- crt.tot – Total length of uninterrupted sequences of capitals
- dollar – Occurrences of the dollar sign, as a percentage of total number of characters
- bang – Occurrences of '!', as a percentage of total number of characters
- money – Occurrences of 'money', as a percentage of total number of characters
- n000 – Occurrences of the string '000', as a percentage of total number of characters
- make – Occurrences of 'make', as a percentage of total number of characters
- yesno – A factor variable indicating if the email was spam, 'y', or not spam, 'n'

**Task**

Imagine you have been asked by your employer to identify features present within spam emails to improve the company spam email filters functionality.

- Which text characteristics influence whether an email will be classified as spam or not?

You should conduct an analysis to answer your question using a Generalised Linear Model (GLM). Following your analyses, you should then summarise your results in the form of a presentation.